REQUIREMENT ANALYSIS

Technology Stack (Architecture & Stack)

Date	03 October 2026
Team ID	SWTID-2026-6390
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Technical Architecture

The whole project is built inside a ServiceNow instance. The architecture below shows only the parts the project actually uses: users, the platform screens, the ACL engine and the data store. No external API, machine learning service or separate database is used.

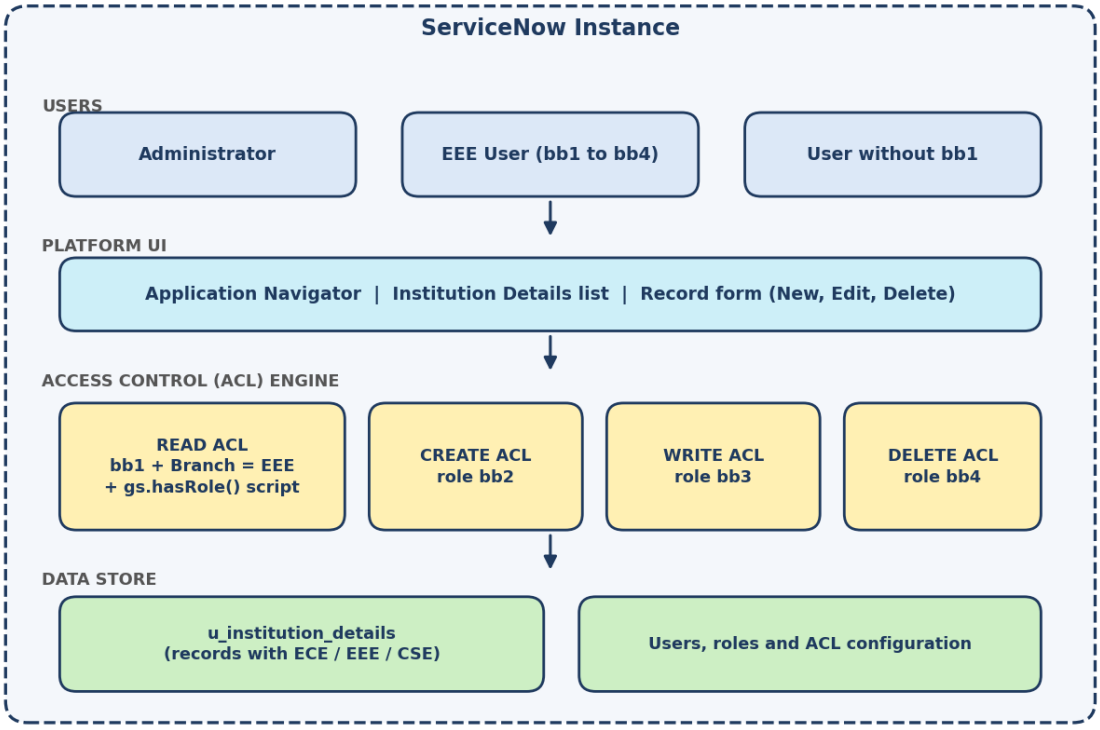


Figure 1: Architecture of the Script-Controlled ACL project.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	The administrator and test users work in the ServiceNow platform screens.	ServiceNow Platform UI
2.	Application Logic-1	Users, roles, the table and the records are created and set up in ServiceNow.	ServiceNow Administration / Configuration
3.	Application Logic-2	The READ ACL checks the role, the Branch condition and the script.	ServiceNow ACL + GlideSystem (gs) script
4.	Application Logic-3	CREATE, WRITE and DELETE are controlled by separate ACLs that need bb2, bb3 and bb4.	ServiceNow Access Control List
5.	Database	Holds the custom table and its records, along with users, roles and configuration.	ServiceNow Platform Data Store
6.	Cloud Database	No separate cloud database is used.	Not used
7.	File Storage	No file storage is used.	Not used
8.	External API-1	No external API is used.	Not used
9.	External API-2	No second external API is used.	Not used
10.	Machine Learning Model	No machine learning model is used.	Not used
11.	Infrastructure (Server / Cloud)	The project is built and tested on a ServiceNow instance.	ServiceNow Instance

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is used.	Not used
2.	Security Implementations	Record-level ACLs use roles and, for READ, the Branch = EEE condition and a script. The administrator is allowed full READ access.	ServiceNow ACLs; roles; data condition; GlideSystem script
3.	Scalable Architecture	No special scalable architecture such as 3-tier or microservices is set up.	Not used
4.	Availability	No load balancer or redundancy setup is used.	Not used
5.	Performance	No caching, CDN or response-time target is set.	Not used